

ROLE-ALIGNED IDENTITY

Make complex systems become reliable work.

Operator-technologist who connects AI, data, engineering, controls, and adoption into governed operating mechanisms. Direct experience spans AI-enabled operations, high-reliability engineering, enterprise workflows, and scaled execution.

Experience selected for the mandate

Vape-Jet - Director of Operations | 2025-Present

- Leads AI-first operating transformation across production, purchasing, quality, customer success, support, engineering issue flow, ERP/Odoo, Jira, HubSpot, Slack, telemetry, and AI-assisted workflows.
- Connects factory, field, customer, and machine-fleet signals into prioritized work, earlier risk detection, cleaner handoffs, dashboards, standard work, training, and accountability.
- Works directly in a robotics and machine-vision environment with approximately 1,000 fielded robots informing support and operating feedback loops.

DudeWorth - Founder / AI and Automation Advisor | 2017-Present

- Develops AI-readiness and opportunity frameworks assessing value, feasibility, data readiness, governance, adoption burden, reuse, and time-to-value.
- Designs agentic workflow patterns using structured context, retrieval, AI agents, human judgment, escalation, decision records, evaluation, and governance rails.
- Advises on AI-enabled operations, logistics technology, warehouse automation, AMRs, ASRS, cobots, transportation systems, network analysis, and workflow design.

Amazon - Operations Management / Site Leader | 2014-2018

- Led launch and execution for Amazon Prime Pittsburgh in a 24/7 customer-backward environment supporting approximately five million second-day orders annually.
- Built leadership routines, labor planning, throughput visibility, escalation paths, standard work, and operating mechanisms.
- Applied Lean and Gemba practices across safety, quality, throughput, customer experience, and team engagement; contributed to Prime Air research and Amazon Flex geolocation concepts.

Compunetics - Director of Operations and Engineering Management | 2000-2013

- Progressed through R&D; program management, production management, strategic acquisition, operations management, and director-level leadership.
- Led engineering, manufacturing, quality, customer delivery, and complex technical programs in advanced electronics and high-reliability environments.
- Built quality systems, standard work, root-cause practices, performance measures, and process controls; collaborated with DoD, CERN, Intel, and AMD.
- Published industry articles, delivered technical talks, and participated in the HyperTransport Consortium.

Role-aligned capabilities

Enterprise AI patterns	RAG and agent workflows
Data and workflow governance	Human-in-the-loop design
Architecture review mechanisms	Observability and escalation
Cost and value framing	API / system integration
Risk and quality systems	Cross-functional influence

Architecture posture

Modular at the edge. Preserve scientific and workflow context.

Load-bearing at the core. Standardize authority, controls, integration, observability, economics, and lifecycle ownership.

Instrumented end to end. Make performance, risk, cost, adoption, and learning visible.

Current toolkit

Agentic AI · LLM workflows · RAG patterns · human-in-the-loop design · Python · SQL · JavaScript · TensorFlow · Tableau · ERP/Odoo · Jira · Confluence · HubSpot · Slack · GIS and mapping workflows

Decision lens

Standardize / localize

Which patterns become the paved road, and where does scientific context justify bounded variation?

Promote / hold

Which evidence warrants enterprise reuse, and which authority, risk, cost, or ownership gap should stop scale?

Evidence supporting the architecture posture

Build

Agentic workflow patterns, RAG concepts, structured context, human review, evaluation, and governance rails.

Integrate

ERP, CRM, telemetry, production, quality, support, purchasing, logistics, and engineering issue flows.

Assure

Quality systems, root cause, process controls, regulated operations, safety, and chain of custody.

Scale

24/7 customer-backward operations, distributed execution, and a fielded robotics feedback environment.

Claim boundary

Does not claim prior formal enterprise AI architect title or production ownership of Thermo Fisher's exact AWS/Azure, Databricks, Dataiku, UiPath, or vector-database estate. Proposes proving fit through one bounded reference pattern with incumbent specialists.

COMPLEMENTARY SCALE, ASSURANCE, AND TRANSFER

Evidence that transfers across the architecture boundary.

Complementary experience

ZeusVu - Chief Pilot, Autonomous Systems and Aerial Data Operations | 2018-2022

- Led regulated drone operations across medical, energy, logistics, real-estate, and restricted-airspace contexts with a 100% safety record and no FAA incidents.
- Developed TensorFlow-based concepts for automated aerial inventory and computer-vision-enabled field intelligence.
- Designed medical drone delivery and chain-of-custody concepts connecting technology, risk, stakeholders, and governed execution.

Cardinal Building Products - Regional Director | 2021-2023

- Built the company's first remote regional operation.
- Modernized distributed customer, CRM, commercial, fulfillment, logistics, digital marketing, SEO, and online-buying workflows.
- Advanced AI-enhanced transportation-management and route-planning concepts.

Enviro Social Capital - Chairman of the Board | 2014-2017

- Co-founded and led a public-private impact initiative focused on sustainable infrastructure, green finance, and civic transformation.
- Helped shape and evaluate a \$200 million Social Impact Bond feasibility concept for Pittsburgh sewage and green-infrastructure improvement.

IEEE - CMPT and EDS Pittsburgh Chapter Chair | 2012-2016

- Led regional technology programming connecting industry, academia, STEM outreach, electronics, manufacturing, and emerging technology.
- Organized conferences, seminars, business tours, and technical programming.

Transfer map

AI capability design
Agentic AI, LLM workflows, RAG patterns, structured context, human review, escalation, decision records, evaluation, and governance rails.

Data and integration
ERP, CRM, telemetry, production, quality, support, purchasing, logistics, and engineering issue flows.

Reliability and assurance
Quality systems, root cause, process controls, regulated autonomy, safety, chain of custody, and high-consequence technical execution.

Enterprise adoption
Operating cadence, standard work, training, visibility, cross-functional translation, and distributed execution.

Education

University of Pittsburgh
Bachelor of Science
Academic grounding includes materials science, physical chemistry, and biology-related studies.

Credentials

Google AI Essentials
IBM Enterprise Design Thinking Practitioner
Six Sigma Certification
OSHA 10

Operating mechanisms I bring to the architecture role

Mechanism	Verified evidence	Architecture relevance
Operating visibility	Throughput visibility, dashboards, machine-fleet and customer signals, and performance measures.	Convert fragmented activity into a shared operating truth.
Decision cadence	Leadership routines, prioritized work, escalation paths, standard work, and accountability.	Move recurring architecture decisions out of improvisation without removing judgment.
Quality and learning	Quality systems, root-cause practices, process controls, and earlier risk detection.	Turn exceptions and incidents into control changes and reusable patterns.
Distributed execution	Remote regional operations plus CRM, commercial, fulfillment, logistics, and digital workflows.	Support federated teams with clearer interfaces, ownership, and adoption.
AI workflow design	Readiness frameworks, agentic patterns, human review, evaluation, governance, adoption, and reuse.	Attach AI to work, authority, evidence, economics, and sustainable ownership.
Architecture judgment Translate between outcome, data authority, AI behavior, human authority, control, integration, operations, economics, and reuse.		Scientific trust Preserve domain context while applying quality systems, traceability, regulated execution, and explicit decision boundaries.